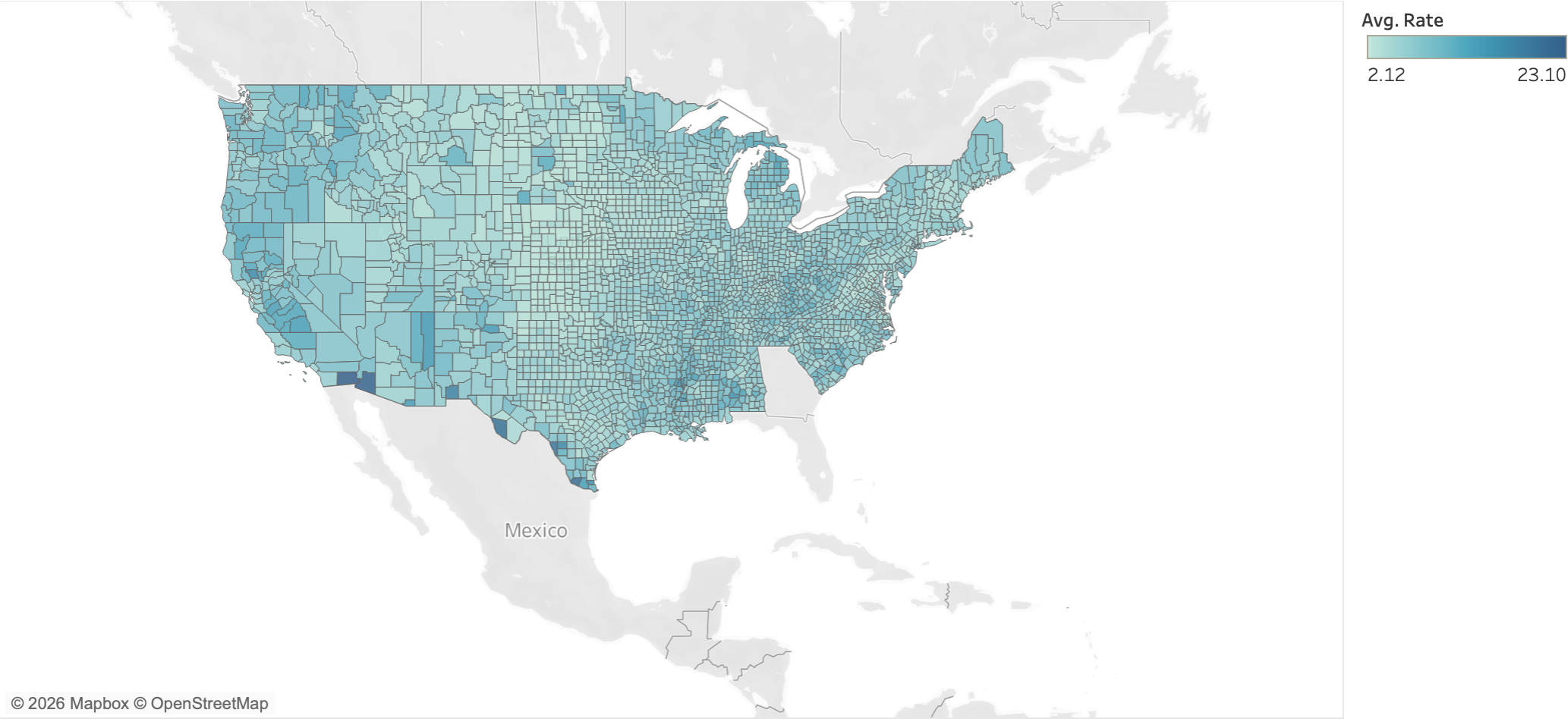


Module4

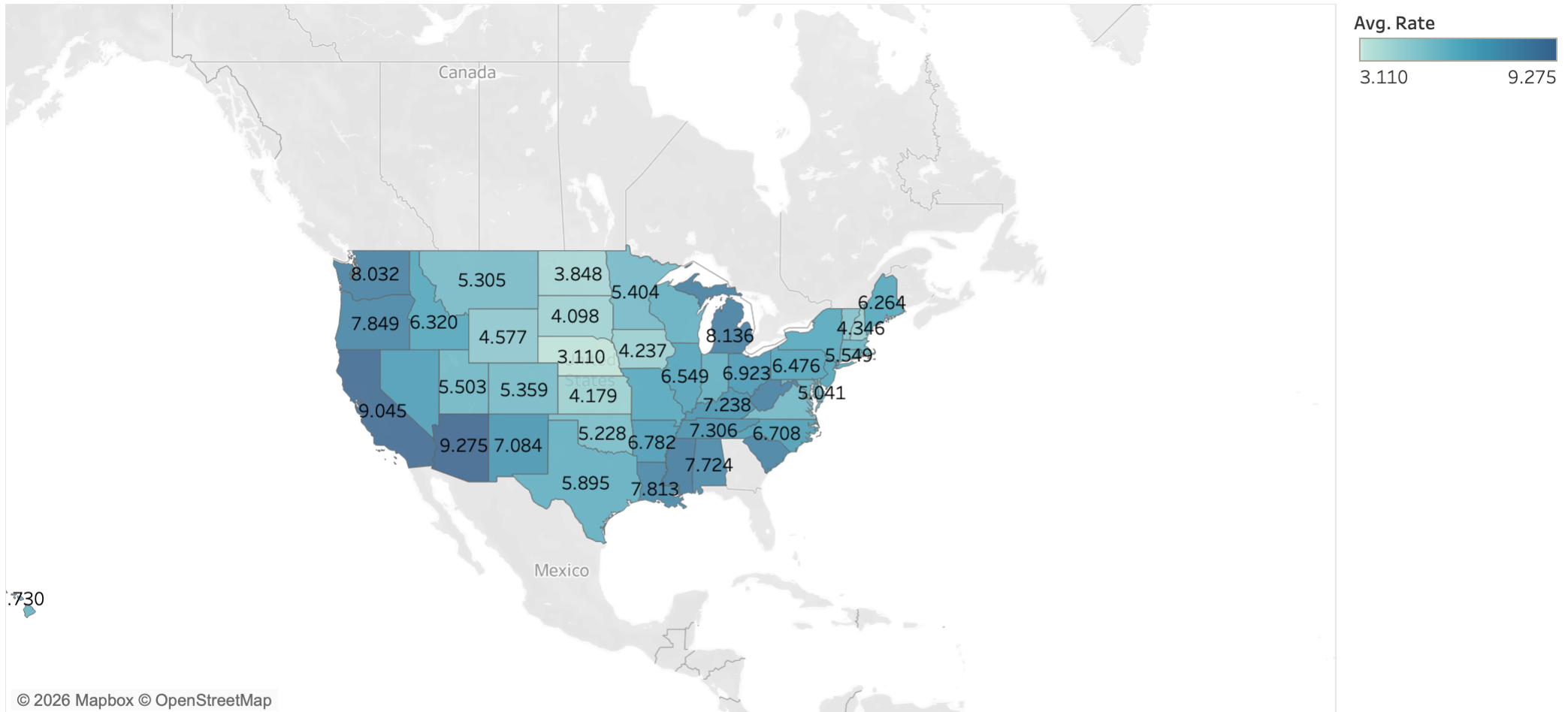
File created on: 7/27/26 1:29:56 PM PDT

US Geographic Unemployment by State County



Map based on Longitude (generated) and Latitude (generated). Color shows average of Rate. Details are shown for County and State. The data is filtered on Year, which ranges from 1990 to 2016.

US Geographic Unemployment by State

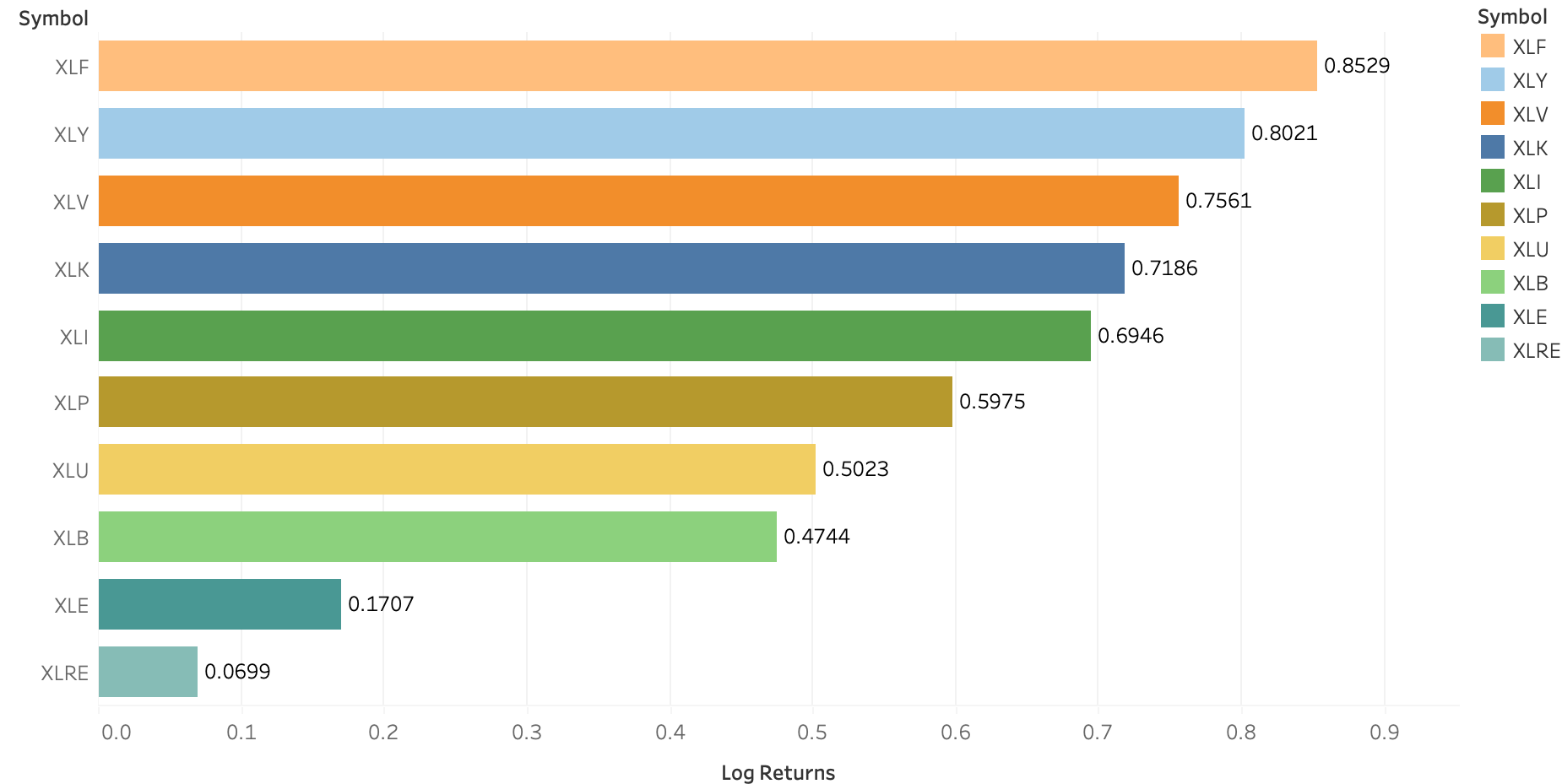


Question: Which states have the highest average unemployment rate?

Answer: The state with the highest unemployment rate is Arizona at 9.275%. California has the second highest rate at 9.045%.

Map based on Longitude and Latitude. Color shows average of Rate. The marks are labeled by average of Rate. Details are shown for State. The data is filtered on Year, which ranges from 1990 to 2016.

Comparative Differences Return of Sector

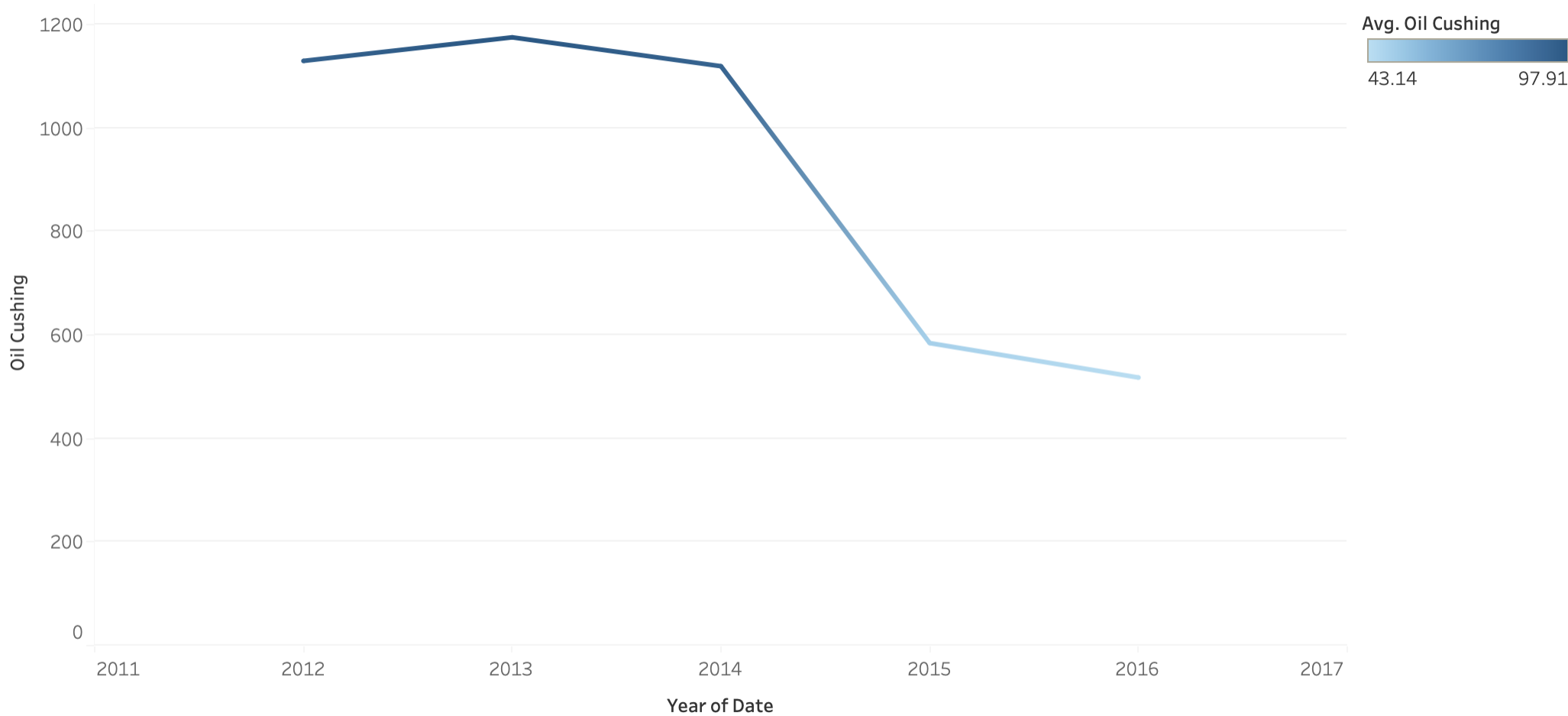


Sum of Log Returns for each Symbol. Color shows details about Symbol. The marks are labeled by sum of Log Returns. The data is filtered on Date1 Year and Year. The Date1 Year filter ranges from 2012 to 2023. The Year filter ranges from 1990 to 2016. The view is filtered on Symbol, which keeps 11 members.

Year Over Year Unemployment by State (Average 2007-2009)

State	Year																											Average Recession	
	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016		
Alabama	8.23	9.08	9.30	9.18	7.62	7.61	7.24	6.92	6.17	6.62	5.50	6.25	7.03	7.08	6.67	5.18	4.77	4.90	6.98	13.17	12.36	11.21	9.26	8.51	7.92	7.03	6.77	3.443	9.551
Alaska	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.347	
Arizona	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.802	
Arkansas	7.64	8.30	7.97	6.96	6.18	5.67	6.43	6.32	6.30	5.62	4.84	5.65	6.20	6.74	6.53	6.09	5.89	6.05	6.21	8.66	9.10	9.30	8.53	8.30	6.97	6.02	4.66		
California	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.974	
Colorado	8.30	8.36	12.03	12.01	10.86	10.61	9.93	9.16	8.77	7.50	6.27	6.62	7.60	7.93	7.53	6.82	6.28	6.80	8.65	12.42	13.89	13.33	11.64	10.38	8.82	7.56	6.92		
Connecticut	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.291	
Delaware	5.26	6.05	7.47	6.21	4.94	4.82	5.23	4.52	4.75	4.10	3.30	3.90	5.02	5.55	5.20	4.92	4.22	3.58	4.62	6.52	8.61	8.28	7.88	7.09	5.20	4.03	3.42		
District of Columbia	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	4.908	
Florida	5.15	6.59	7.48	6.61	5.55	5.43	5.46	5.00	3.45	2.79	2.31	2.98	4.15	5.20	4.81	4.58	4.76	4.26	4.38	5.54	7.72	8.89	8.55	8.13	7.55	6.44	5.49	5.13	
Georgia	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	5.881	
Hawaii	4.38	5.91	5.30	5.25	5.00	4.46	4.97	4.07	3.83	3.68	3.85	3.57	3.94	4.15	3.80	3.93	3.43	3.40	4.97	8.23	8.56	7.73	7.40	6.93	5.90	5.04	4.43		
Idaho	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	5.532	
Illinois	3.64	4.14	7.12	7.45	8.36	8.41	8.55	8.32	7.73	6.51	4.47	4.80	4.39	4.26	3.54	2.94	2.68	2.96	4.90	8.79	8.27	8.08	6.86	5.46	4.75	3.93	3.41		
Indiana	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	5.549	
Iowa	6.32	7.31	7.53	7.36	6.48	6.37	6.41	6.33	6.32	6.13	5.62	5.94	6.26	6.38	5.74	4.69	4.11	3.44	5.37	8.80	9.53	9.00	7.86	6.84	5.43	4.70	4.37		
Kansas	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	5.072	
Kentucky	6.68	7.38	7.45	5.98	5.76	5.76	5.01	6.62	4.02	4.56	4.45	4.56	5.30	5.90	6.08	5.44	4.71	5.17	6.60	9.94	9.08	8.55	8.91	6.95	6.16	6.23			
Louisiana	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	7.239	
Maine	6.02	6.73	7.15	5.83	5.28	5.02	4.57	4.00	3.46	3.39	3.33	4.40	5.32	5.48	5.56	5.71	5.22	4.71	6.15	10.91	10.71	9.29	8.45	7.75	5.96	4.81	4.62		
Maryland	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	7.258	
Massachusetts	4.36	4.71	4.54	4.18	3.77	3.77	3.81	3.33	3.01	2.80	2.79	3.49	4.10	4.66	4.57	4.42	3.83	3.85	4.39	6.79	6.13	5.57	5.03	4.74	4.21	3.68	3.88		
Michigan	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	5.011	
Minnesota	3.44	3.67	3.85	4.19	4.21	3.96	3.92	3.62	3.36	3.04	3.31	3.88	4.23	4.43	4.71	4.38	3.82	3.80	4.01	5.66	6.01	5.62	5.05	4.69	4.02	3.96	3.98		
Mississippi	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	4.494	
Missouri	7.27	8.98	8.32	7.53	6.52	6.68	6.83	6.69	5.91	5.80	4.94	5.92	6.32	6.87	6.08	6.57	6.39	6.14	7.18										
Montana	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	6.345	
Nebraska	6.10	8.40	8.99	9.10	8.30		7.34	7.48		6.73	6.08				6.45	6.92	7.63			5.89	7.65	9.02	8.93	8.05	7.67	7.24	7.37	7.22	
Nevada	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	6.474	
New Hampshire	5.73	7.99	7.82	7.86	7.69	6.69	5.96	5.95	5.33	4.80	3.86	4.33	4.86	5.57	5.20	5.55	5.19	5.35	6.21	8.98	8.73	8.89	8.19	7.30	6.24	4.82	4.24		
New Jersey	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	6.845	
New Mexico	5.72	7.44	8.22	7.24	6.23	6.05	6.27	6.09	5.36	4.31	3.92	4.33	4.62	4.71	4.48	4.36	4.14	3.80	4.67	7.66	8.50	8.10	7.73	7.30	6.44	5.78	4.91		
New York	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	5.378	
North Carolina	6.21	8.69	8.51	7.13	6.21	5.41	4.64	4.17	3.52	3.33	2.73	3.54	4.78	5.27	4.88	4.68	4.74	4.50	5.43	8.02	8.79	7.81	7.19	7.23	6.23	5.43	4.14		
North Dakota	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	5.986	
Ohio	8.86	10.45	10.67	9.23	8.51	7.76	6.85	5.96	5.21	4.96	4.74	6.39	7.26	8.02	7.84	7.98	8.07	8.69	13.67	13.35	11.38	10.12	9.98	8.33	5.94	5.57			
Oklahoma	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	8.534	
Oregon	6.09	6.44																											

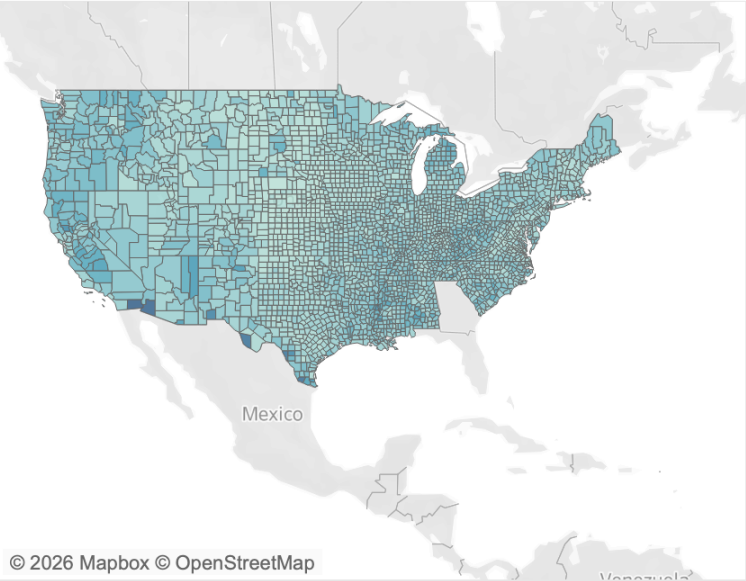
Question 5: Change in Oil Price by Year



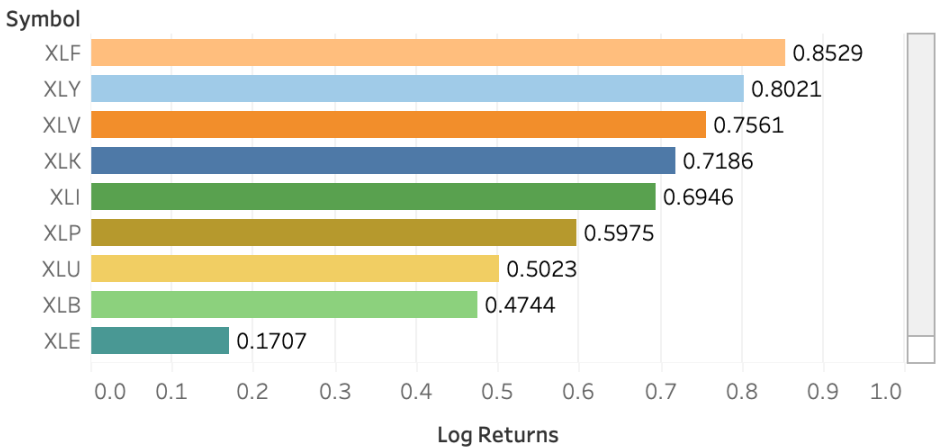
The trend of sum of Oil Cushing for Date Year. Color shows average of Oil Cushing. The data is filtered on Year (oil!cushing) and Year. The Year (oil!cushing) filter ranges from 2012 to 2023. The Year filter ranges from 1990 to 2016.

Geospatial Analysis and Network Analysis

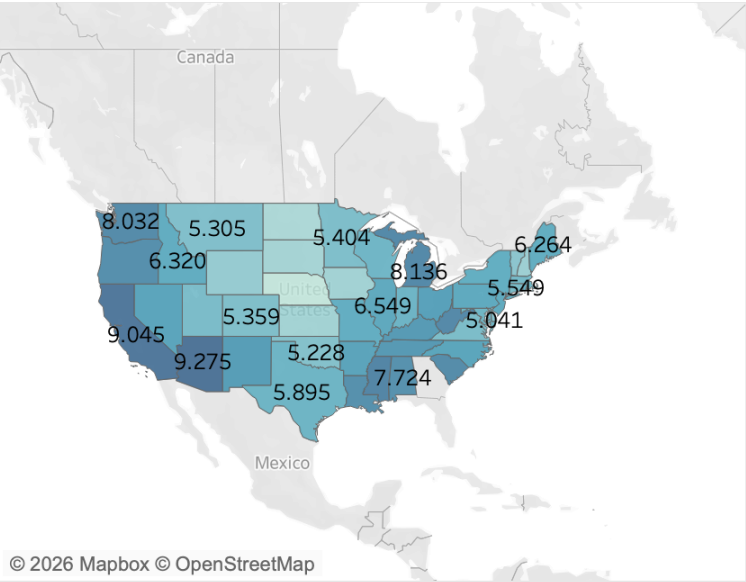
US Geographic Unemployment by State County



Comparative Differences Return of Sector



US Geographic Unemployment by State



Year Over Year Unemployment by State (Average 2007-2009)

State	Year							
	1990	1991	1992	1993	1994	1995	1996	1997
Alabama	8.23	9.08	9.30	9.18	7.62	7.61	7.24	6.9
Arizona	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.34
Arkansas	8.29	8.94	11.74	11.46	11.45	10.65	11.65	9.9
California	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.80
Colorado	7.64	8.30	7.97	6.96	6.18	5.67	6.43	6.3
Florida	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.97
Georgia	8.30	8.36	12.03	12.01	10.86	10.61	9.93	9.1
Idaho	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.29
Illinois	5.26	6.05	7.47	6.21	4.94	4.82	5.23	4.5



Year
1990 to 2016